PROTOCOL

Methods to select outcomes in Core Outcome Set development: protocol for an expert Vignette-survey

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Gregor Stolte designed the study, extracted and analyzed the data and wrote the first draft.
Jérôme Lang, Jean-François Laslier and Viet-Thi Tran designed the study, critically edited the manuscript.
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Introduction

Epidemiological data suggests that between 6% and 8% of 18-24 year olds globally have shown symptoms of a depressive episode within the last 12 months (Hermann et al., 2022). In conjunction with the fact that depression is the second leading cause of years lived disability, it is a major public health problem that requires preventative and therapeutic interventions for large populations (GBD Mental Disorders Collaborators., 2022). The development of these interventions, as well as their market authorizations, reimbursement decisions, and recommendations for good clinical practice by health authorities, are determined by their effectiveness on depressive symptoms and acceptable levels of adverse effects.

These decisions are based on observing the effect of therapeutic interventions in key outcomes in randomized controlled trials [RCTs], which are defined by a varying number of domains [e.g., anxiety, depression severity, etc.] (Zarin et al., 2011). In depression, the past 10 years of research have been difficult to compare due to a large heterogeneity of outcome judgment criteria for demonstrating treatment effectiveness, resulting in a waste of research (Williamson et al., 2012; Chalmers et al., 2014). Current RCTs are designed on the basis of non-generalizable and rarely patient-centered explanatory trials (Barbui & Patten, 2018), leading to the potential that the outcomes included are not relevant to patients. The inclusion of heterogeneous outcomes across trials limits the generalizability of the meta-analysis results due to a lack of overlap in judgment criteria (Chalmers et al., 2014; Williamson et al., 2012). Indeed, two studies investigating measurement overlap of depression outcomes concluded that there was a need to develop common outcomes for better comparability and generalizability (Fried, 2017; Santor et al., 2006).

A promising method to address this challenge is developing a Core Outcome Set [COS] for depression interventions and treatments that could improve the quality of future RCTs (Chevance et al., 2022; Williamson et al., 2017), and eventually be used in meta-analyses (Williamson et al., 2014). A Core Outcome Set is defined as a minimum set of outcomes that all relevant stakeholders agree upon prior to its use in trials (Comet Handbook, 2017; Williamson et al., 2012). In addition to limiting heterogeneity, the development of a COS for depression could also improve the comparison, analysis, and evidence synthesis of heterogeneous outcomes (Williamson et al., 2012). Furthermore, international methodological recommendations for conducting RCTs [as is the case for depression treatments] support the necessity to use a COS when one is available (Moher et al., 2010).

The first step of a COS is to identify what to measure, usually represented by a long-list of important outcomes. This list is then refined through consensus methods for which outcomes are selected by relevant stakeholders to be included in the COS (COMET Handbook, 2017). Selecting outcomes is a crucial step in COS development and while it can improve the validity and usefulness of RCTs, it has the ultimate goal of improving the lives of people with depression (Coster, 2013). One commonly recommended method for selecting outcomes in a COS is the Delphi method (Beaton et al., 2021; COMET Handbook, 2017). This method consists of several rounds where participants evaluate the importance of an outcome on a criterion from 1 to 9. Each participant then receives feedback about how others evaluated the outcome and is invited to revise their evaluation. Over several rounds, a consensus is reached and guidelines on consensually important outcomes are established. However, this process has several limitations. First, while international recommendations for the development of a COS underline the necessity of including both patients and clinicians (Beaton et al., 2021; Boers et al., 2014; Chevance et al., 2020; COMET Handbook, 2017; Kirkham & Williamson., 2022), outcome selection studies often have different proportions of patients and experts involved in the determination of criteria. A study by Chevance et al. (2020) determined that out of 63 participants, only 12% were patients. The low involvement of patients and the public in COS development studies increases the probability that the outcomes generated and selected are not relevant to patients. Additionally, the way in which participants are encouraged to compromise their beliefs as to what is important across several rounds of a Delphi study (Donohoe & Needham., 2009) severely impacts the "true selection" of important outcomes. When participants are encouraged to modify their choices to more closely align with others, there is the risk that the chosen outcomes do not represent what is truly important. Hence, there is a clear need to identify more effective methods for selecting important outcomes to include in a COS.

Objectives

The main objective of this thesis is to identify adequate methods to elicit and aggregate patients' preferences toward outcomes in the context of COS development.

Taking the example of the development of a COS for depression, this work builds on a preliminary work that identified 80 outcome domains that matter to patients with depression (Chevance et al., 2020). This thesis will focus on identifying the most appropriate methods to elicit individual patients' preferences toward this long-list of outcomes and aggregate them into a collective preference.

Methods

The protocol of the survey was registered on Zenodo before the inclusion of the first participant.

Design

Using an online expert survey based on experimental vignettes methodology, and disseminated among experts, we will determine the most appropriate methods to elicit and aggregate patients' preferences toward outcomes in the context of COS development.

We will use the 'Paper People Studies' which are a specific type of Experimental Vignettes Methodology [EVM], typically presented in a written form.

Vignettes are short hypothetical accounts reflecting real-world situations (Tremblay et al., 2022). Participants are asked to make explicit decisions and judgments about the proposed methods and justify them (Aguinis & Bradley., 2014). It is advantageous to use vignettes in the survey due to their hybrid methodology between survey research and experimental methods (Atzmueller & Steiner., 2010; Evans et al., 2015). In particular, vignettes allow for the collection of qualitative and quantitative data that supports the experts' opinions. Vignette-based Delphi studies are typically used to identify solutions in situations of great uncertainty, based on the judgment of different experts. For instance, they have been used among health researchers to generate reasonable solutions to ethical dilemmas (Bromley et al., 2017). Vignettes have also been used to determine the best study designs for assessing interventions to improve the peer review process and to determine whether the study designs used were the most appropriate (Heim et al., 2018).

Population - eligibility criteria

Inclusion criteria

- Academic [post-doc and above] in social choice theory, behavioral economics, political sciences, health sciences or marketing, with expertise in preference elicitation methods or aggregation methods

Exclusion criteria

- None

Recruitment and Dissemination of the survey

This study relies on voluntary response sampling. Recruitment of participants will be achieved through three contact methods: :

- International experts in preference elicitation/aggregation will be contacted via the mailing list of the 17th World Congress of the Society for Social Choice and the Measurement of Well-Being [~150 participants potentially meeting the eligibility criteria are expected].
- Institutional and academic mailing lists of the authors of the paper with expected snowball effect of the relevant departments: social choice theory, behavioral economics, political sciences, health sciences or marketing.
- The constitution of a mailing list of experts identified through publications of the last 2 years identified by google scholar. These publications encompass theoretical and empirical work on elicitation and/or aggregation methods in the following fields: social choice theory, behavioral economics, political sciences, health sciences or marketing.

Reminder emails will be sent once a month to encourage more responses.

Population - Sample Size

There is no officially determined sample size for vignettes surveys as each approach has its own prerequisites. Previous research using vignettes in combination with experts did not differ in sample size [e.g., 224, 343] due to the individual nature of each research method (Bakker, Jolly, et al., 2014; Bakker, Edwards, et al., 2014; Heim et al., 2018). Despite this, most of the aforementioned studies tested more vignettes than the current research, justifying a larger sample size. We set a lower threshold of at least 50 participants.

Development of the survey and collected data

The survey was developed iteratively by all authors with different expertise in social choice theory, behavioral economics, clinical epidemiology, and public health.

The format of the questionnaire was developed according to theories of survey design (Tremblay et al., 2022). The survey will be anonymous, meaning that no personal or identifying data will be collected from participants.

To limit forcing responses, participants will be provided the possibility to refrain from answering questions where they do not believe they are experts in the subject matter (Bakker et al., 2014).

We will use likert scales ranging from 1-4 as it simplifies the task of responding to each question and is less burdensome. Eliminating the neutrality option encourages participants to actively choose the most appropriate methods and will help us make a decision.

The survey will be presented in four sections:

- Demographics: This section consists of a few closed-ended questions to assess the eligibility of participants and describe the included sample. For eligible participants, data will be collected on: age, gender, country of residence, research field, work affiliation, experience working after PHD, and experience with preference elicitation/aggregation studies.
- Preference elicitation methods: his section will be presented in a progressive way, consisting of 3 subsets of questions.
  - Subset 1 introduces and describes the constraints associated with 5 families of consensus methods [Consensus, Choice-based, Ranking, Rating, Trade-off]. Participants are asked to rate whether they believe each family is suited to the task of eliciting patient preferences over 80 outcomes; Likert scale [1-4, Not at all – Yes definitely; option of not responding – I don’t know this method]. These

    - 5 families of methods were identified by a scoping review of preference elicitation methods by Ribeiro et al. (unpublished). Participants have the possibility to justify their answer with free text responses.
    - Subset 2 presents a two-step method developed by the research team and asks participants to rate the adequacy of this proposed method.
    - Subset 3 asks participants what they would finally choose for a method, considering all above-mentioned possible methods and potentially forgotten ones.
- Aggregation methods: This section follows a similar structure to the section on Preference elicitation methods.
    - Subset 1 invites participants to rate four different families of aggregation methods based on their familiarity with the method and suitability for COS development [Best-k method, Proportional method, Diversity focused method, Majoritarian method] on a Likert scale [1-4, Not at all – Yes definitely; option of not responding – I don't know this method]. The families of methods were previously identified by a scoping review of Ribeiro et al. (unpublished). Participants have the possibility to justify their answer with free text responses.
    - Subset 2 presents a desired method of the research team [Ordered-weighted average with harmonic weights method] and asks the participants whether they consider it as adequate to aggregate preferences in the selection of outcomes for a COS.
    - Subset 3 asks participants what they would finally choose for a method, considering all above-mentioned possible methods and potentially forgotten ones
- Complementary questions: This section explores alternative scenarios to those presented in the above vignettes.
    - First, the participant states how they would modify the proposed method of COS development (elicitation and/or aggregation) if the final objective changes the number of core outcomes considered: 25 out of 80 outcomes, instead of 10 out of 80.
    - Second, the participant answers a question regarding the current use of Delphi for developing COS methods and how appropriate the participant believes this method accurately captures patient preferences.
    - Third, the participant answers a question regarding the separability of preferences between outcomes.
    - Fourth, the participant is asked to recommend five scholars whose expert answers to this survey they would trust the most.
    - Fifth, the participant is provided with the email of the researchers if they would like to be mentioned in the acknowledgements of this paper [participants are invited to send an email to the experimenters for recognition].
    - Finally, participants are given the opportunity to provide any comments, questions, and or feedback on the survey.

A physical version of the questionnaire was tested with experts corresponding to the eligibility criteria of the survey prior to its development as a digital survey. Feedback and improvements were discussed and agreed upon during a final consensus meeting with all authors prior to its publication.

The digital survey was published with the Qualtrics account of Maastricht University and tested on a convenience sample of experts with the eligibility criteria mentioned above.

The full questionnaire is available as appendices.

Planned analysis
In the final analysis we will include participants who will meet the eligibility criteria outlined in questions 1 and 2 and those who would have completed at least one of the open-ended questions of the elicitation or aggregation section [question 3 in each respective section].

- Demographics: Characteristics of participants will be statistically analyzed with frequencies and proportions.
- Elicitation and aggregation methods: All open response questions will be analyzed with a standard qualitative content analysis of the manifest content related to the choice of a method for the development of a COS (Hsieh & Shannon, 2005; Vaismoradi et al., 2013). Then, we will group all arguments for each method gathered by the qualitative content analysis as by their pros and cons. To answer the primary objective, all co-authors of the project will consider 1) whether a method of the subset 3 is reported frequently with convincing arguments, 2) whether one of the proposed methods in subset 2 is identified as adequate by a majority of the participants with convincing arguments and finally consider the answers to subset 1. Discrepancies between the answers will be discussed with all researchers involved in the project to identify one or several methods to be used in the patients' survey.
- Complementary questions: All questions in this section will be analyzed using qualitative content analysis as well as descriptive statistics.

Ethics
According to French Law, this survey does not need the evaluation of an ethical committee. The online vignette survey will not collect personal data beyond the extent of anonymous data required for analysis. The collection of data and their storage will be compliant with the EU GDPR laws. Data will be analyzed by researchers of the CRESS (Dr Astrid Chevance and Gregor Stolte) and of PSE (Antonin Macé)

Expected results
The results of this experiment will identify a core set of preference elicitation and aggregation methods from which a COS can be advised and further developed. Furthermore, the results will provide insight into the objective of determining whether the study designs [Delphi, NGT] used in practice are rated as the most appropriate.

Perspectives
For research
The contribution of this study identifies the most appropriate preference elicitation and aggregation methods from the perspective of experts that could address the development of a COS for depression. Following this study, we will develop a survey to prioritize the 80 candidate outcomes to evaluate the efficacy of treatment for depression as identified by Chevance et al. (2020). This study will inform the final decision of what outcomes should be included in the COS.

For clinical practice
The impact of this paper for clinical research is the possibility of testing a COS constructed with insights from different academic fields. The overlap of these fields and interaction with one another allows for a robust construction of a COS. The use of preference elicitation/aggregation methods deemed appropriate by experts in social choice theory, set of outcomes and methodological review developed by experts in epidemiology, and psychology and economics now permit a practical lab test of the COS. Given that future research applies this COS in a lab setting, it may be useful further down the line to inform policy decisions and the improvement of evaluating treatments/interventions for Depression.

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